The Department of Defense and its relationship with Legislators

PS 699: Independent Study (Spring 2020)

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Contents

Data

In all, we have 8005 letters received by the DoD marked "congressional" since 2003. All of these are from Members of the U.S. Congress.

```
load(here("data/all_contacts.RData"))
# one obs per member per letter for all agencies
d <- all_contacts %>% filter(str_detect(agency, "DOD")) %>% filter(!(last_name == "GRASSLEY"& party ==
# would Grassley be considered a good outlier??
# 2007-2017 Data Points
# NA State that they have been coded but not implemented
```

Top 10 Legislators who wrote to DoD combined

```
# top overall
d %>% count(bioname,party,cqlabel, sort = T) %>% top_n(10, n) %>% kable(caption = "Legislators who wrot
```

Table 1: Legislators who wrote the most letters to DOD

bioname	party	cqlabel	n
McCAIN, John Sidney, III	(R)	(AZ)	116
FORBES, J. Randy	(R)	(VA-04)	86
NELSON, Clarence William (Bill)	(D)	(FL)	84
BURR, Richard M.	(R)	(NC)	75
McCASKILL, Claire	(D)	(MO)	70
TOOMEY, Patrick Joseph	(R)	(PA)	60
WITTMAN, Robert J.	(R)	(VA-01)	60
BOOZMAN, John	(R)	(AR)	59
HUNTER, Duncan Duane	(R)	(CA-50)	59
PELOSI, Nancy	(D)	(CA-08)	59

##FILTER OUT NA??

```
d %>% count(TYPE,sort = T) %>% filter(!(is.na(TYPE))) %>% top_n(10, n) %>% kable(caption = "Most Common
```

Table 2: Most Common Policy Areas

TYPE	n
1	3779
5	148
6	88
0	43
2	35
3	25
4	3

```
##Types of Policy each member dealt with
d %>% count(bioname,party, TYPE, cqlabel, sort = T) %>% top_n(10, n) %>% kable(caption = "Legislators wl
```

Table 3: Legislators who wrote the most letters to DOD

bioname	party	TYPE	cqlabel	n
FORBES, J. Randy	(R)	NA	(VA-04)	58
McCAIN, John Sidney, III	(R)	NA	(AZ)	58
McCASKILL, Claire	(D)	NA	(MO)	52
NELSON, Clarence William (Bill)	(D)	1	(FL)	50
BOOZMAN, John	(R)	1	(AR)	48
McCAIN, John Sidney, III	(R)	1	(AZ)	46
WEBB, James H. (Jim)	(D)	NA	(VA)	45
PELOSI, Nancy	(D)	6	(CA-08)	43
TOOMEY, Patrick Joseph	(R)	1	(PA)	40
BURR, Richard M.	(R)	1	(NC)	38

##COULD ANYTHING BE DONE HERE

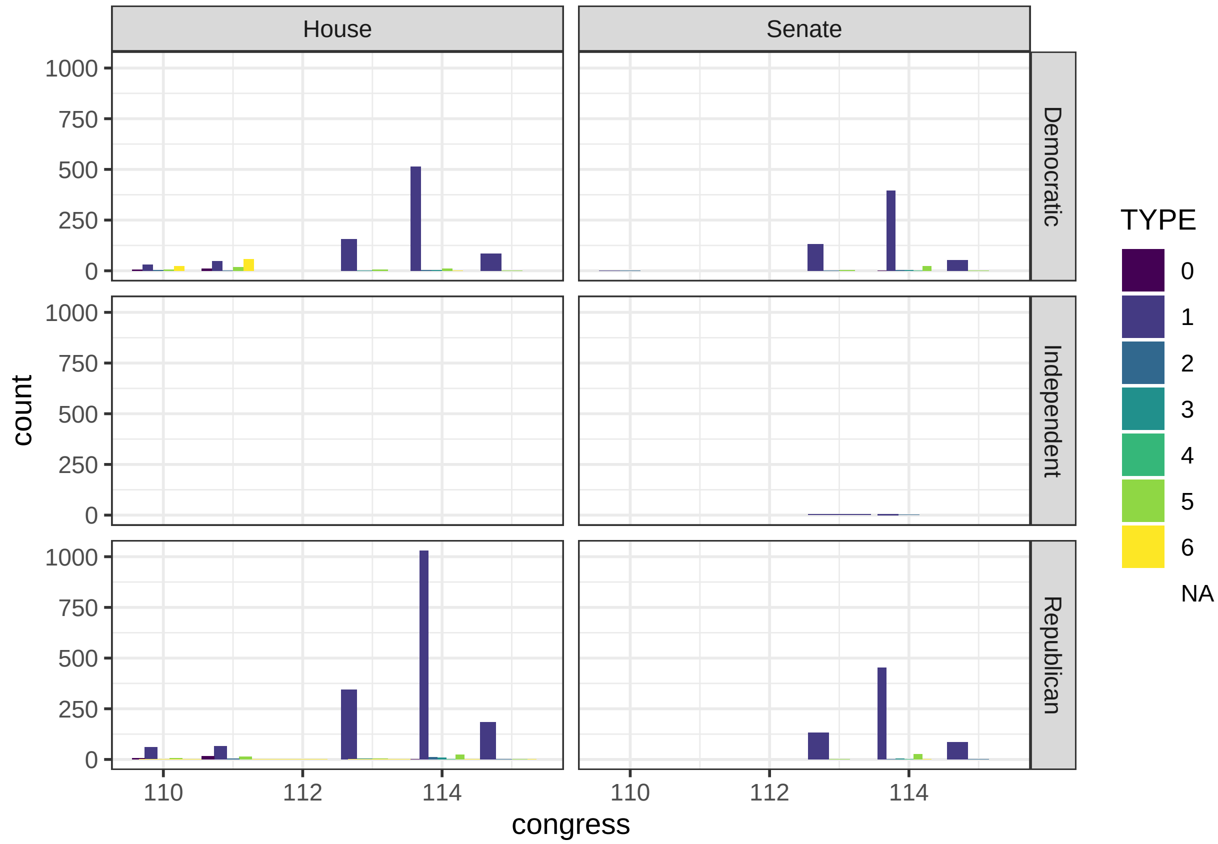
```r
# most with a given position
# d %>% count(bioname, party, cqlabel, letter_position, sort = T) %>% filter(letter_position != "Other"

# most with a given position in a cycle
d %>% count(bioname, party, cqlabel, TYPE, congress, sort = T) %>% filter(TYPE != "Other") %>% top_n(10
```

Table 4: Most frequent letter positions for a given legislator in a cycle

bioname	party	cqlabel	TYPE	congress	n
NELSON, Clarence William (Bill)	(D)	(FL)	1	114	39
BOOZMAN, John	(R)	(AR)	1	114	37
TOOMEY, Patrick Joseph	(R)	(PA)	1	114	32
BURR, Richard M.	(R)	(NC)	1	114	30
PELOSI, Nancy	(D)	(CA-08)	6	111	29
OBEY, David Ross	(D)	(WI-07)	6	111	28
KAINE, Timothy Michael (Tim)	(D)	(VA)	1	114	26
MURRAY, Patty	(D)	(WA)	1	114	25
RIGELL, E. Scott	(R)	(VA-02)	1	114	25
DAVIS, Susan A.	(D)	(CA-53)	1	114	24
McCAIN, John Sidney, III	(R)	(AZ)	1	114	24

```r
#
# # d %>%   filter(str_detect(SUBJECT, "forwards")) %>% add_count(bioname, sort = T) %>% top_n(1, n) %>
#

d %>%
  ggplot() +
  aes(x = congress, fill = TYPE) +
  geom_bar(position = "dodge") +
  facet_grid(party_name ~ chamber)
```

```
## Warning: position_dodge requires non-overlapping x intervals
```



###Senators and Veteran Population ###HELP WITH THIS #Representative per district??

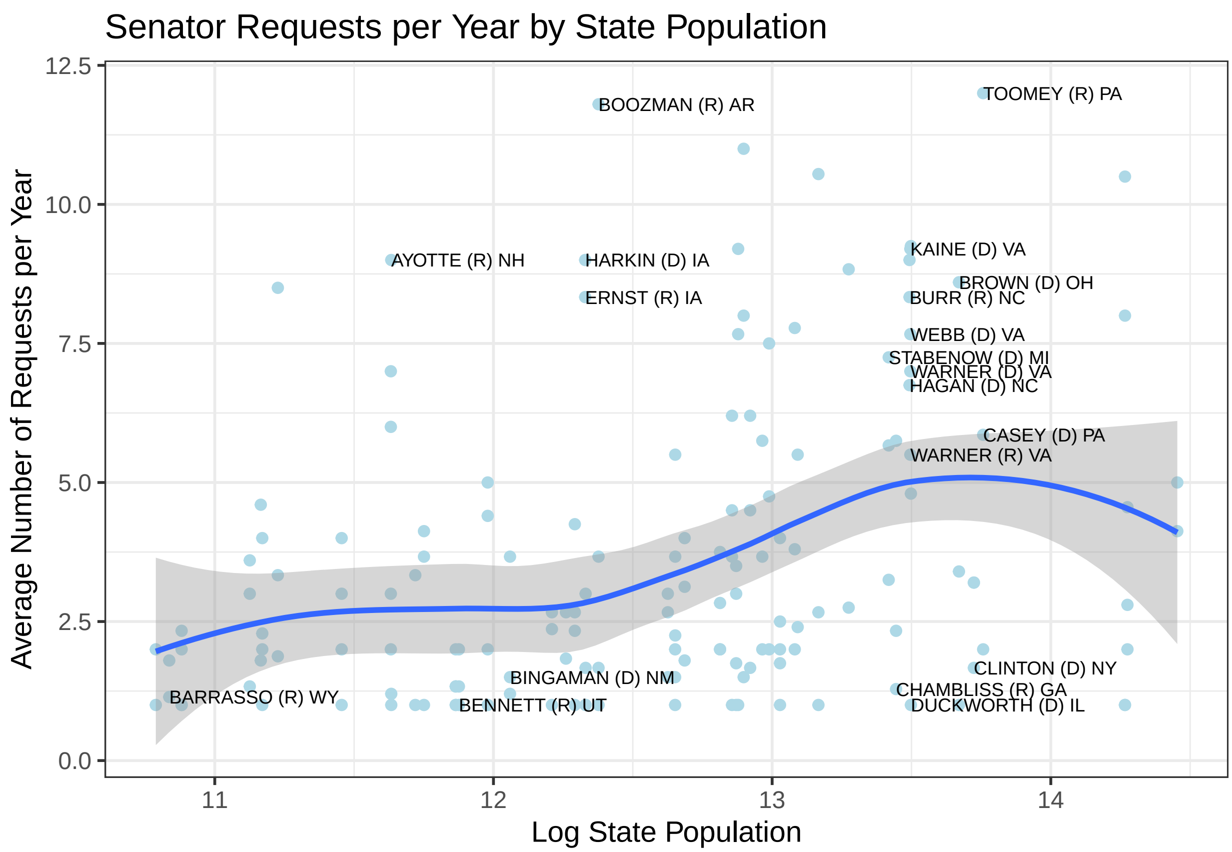
```
vet_states <- read.csv(here("vetpop2010.csv"))
#vet_states <- read.csv("DOD Fatima (Independent Study)/vetpop2010.csv")
vet_states$state %<>% tolower()
vet_states$vetpop2010 <- gsub(",","",vet_states$vetpop2010)
vet_states$vetpop2010 %<>% as.numeric()
#write.csv(vet_states, "DOD Fatima (Independent Study)/vetpop2010.csv")

vet_states %<>% select(state, vetpop2010)
d %<>% left_join(vet_states)
```

```
## Joining, by = "state"
```

```
d %>%
  filter(chamber == "Senate") %>%
    group_by(member_state, year, vetpop2010) %>% summarise(n = n()) %>%
    group_by(member_state, vetpop2010) %>% summarise(mean = mean(n)) %>% ungroup() %>%
  ggplot() +
    geom_point(aes(x = log(vetpop2010), y = mean), color = "light blue") +
  geom_smooth(aes(x = log(vetpop2010), y = mean)) +
  geom_text(aes(x = log(vetpop2010),
                y = mean,
                label = ifelse(mean > mean(mean)*1.5 | mean < mean(mean)*0.5,
                               member_state, "")),
```

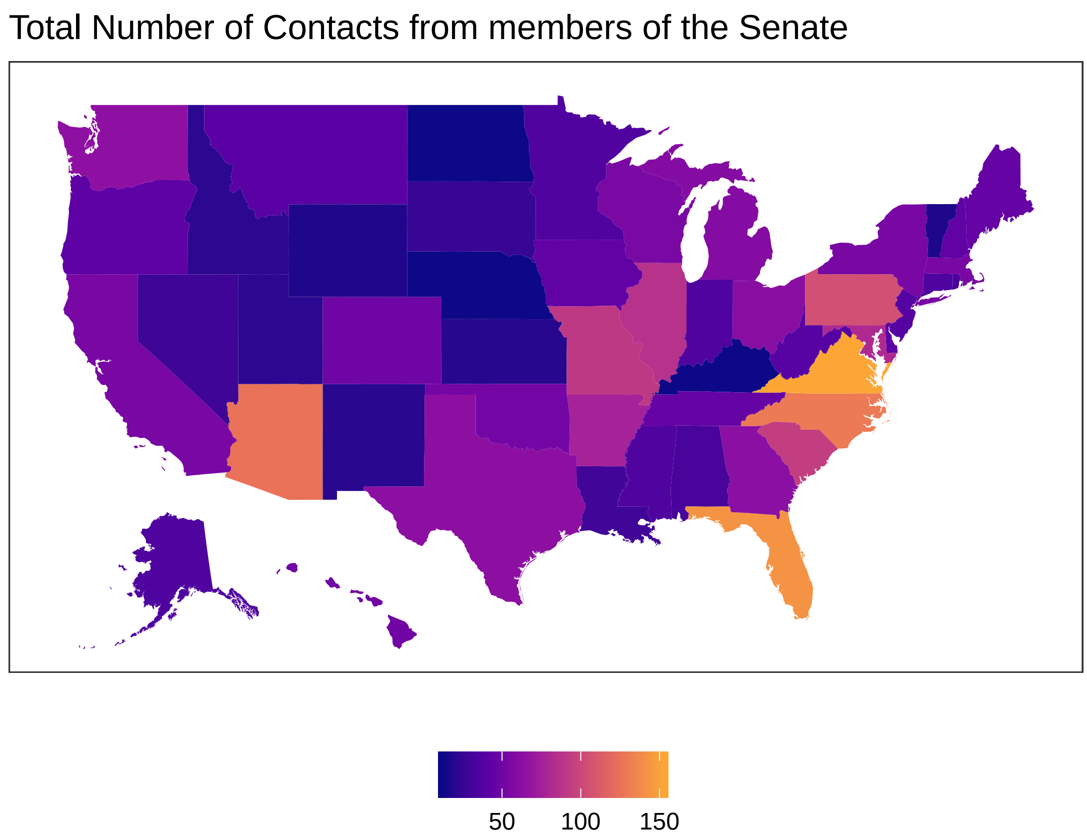
```r
            check_overlap = TRUE,
            size = 2.5,
            hjust = 0) +
  theme_bw() +
  labs(title = "Senator Requests per Year by State Population",
       x = "Log State Population",
       y = "Average Number of Requests per Year")
```

```
## `geom_smooth()` using method = 'loess' and formula 'y ~ x'
```

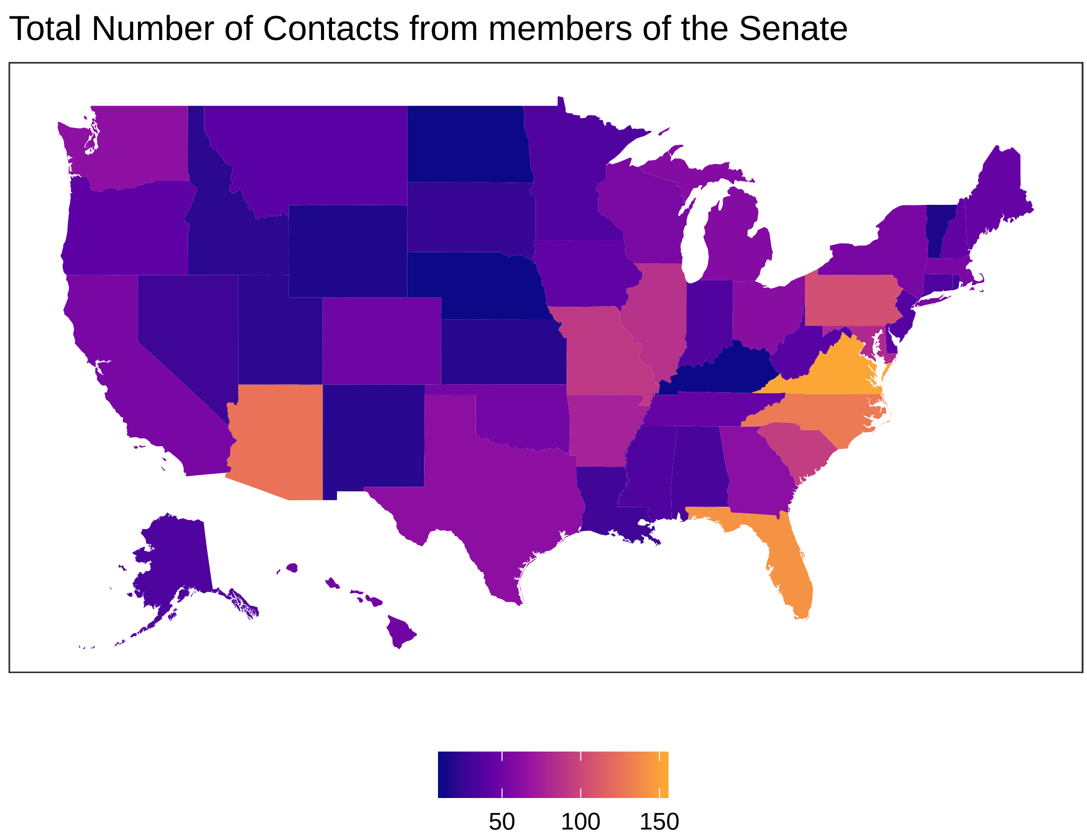



```r
Chamber = "Senate"

d %>%
  ungroup() %>%
  filter(chamber == Chamber) %>%
  group_by(state) %>% summarise(n = n()) %>%
# map_id creates the aesthetic mapping to the state name column
ggplot() +
  # map points to the fifty_states shape data
  geom_map(aes(map_id = state, fill = n), map = fifty_states) +
  expand_limits(x = fifty_states$long, y = fifty_states$lat) +
  coord_map() +
  scale_x_continuous(breaks = NULL) +
  scale_y_continuous(breaks = NULL) +
  labs(x = "", y = "", title = paste("Total Number of Contacts from members of the", Chamber)) +
```

```
  scale_fill_viridis_c(option = "C", end = .8) +
  theme(legend.position = "bottom", legend.title = element_blank(),
        panel.background = element_blank()) # + facet_grid(. ~ Constituent)
```



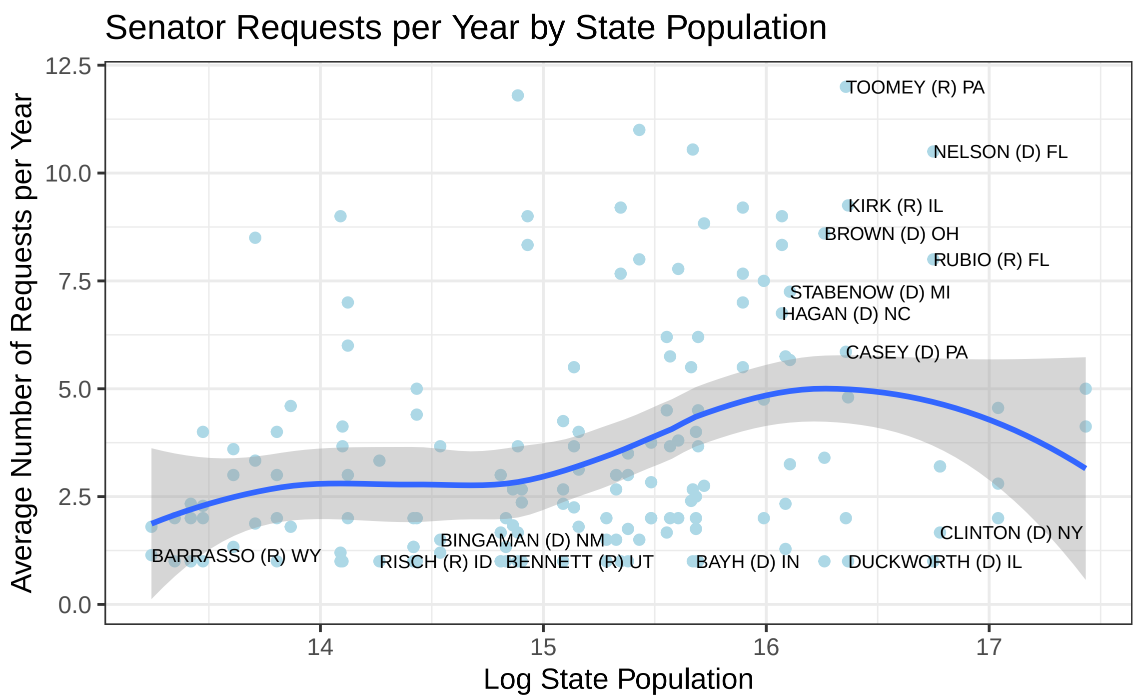

```
Chamber = "Senate"

d %>%
  ungroup() %>%
  filter(chamber == Chamber) %>%
  group_by(state) %>% summarise(n = n()) %>%
# map_id creates the aesthetic mapping to the state name column
ggplot() +
  # map points to the fifty_states shape data
  geom_map(aes(map_id = state, fill = n), map = fifty_states) +
  expand_limits(x = fifty_states$long, y = fifty_states$lat) +
  coord_map() +
  scale_x_continuous(breaks = NULL) +
  scale_y_continuous(breaks = NULL) +
  labs(x = "", y = "", title = paste("Total Number of Contacts from members of the", Chamber)) +
  scale_fill_viridis_c(option = "C", end = .8) +
  theme(legend.position = "bottom", legend.title = element_blank(),
        panel.background = element_blank()) # + facet_grid(. ~ Constituent)
```



Senators vs. state population

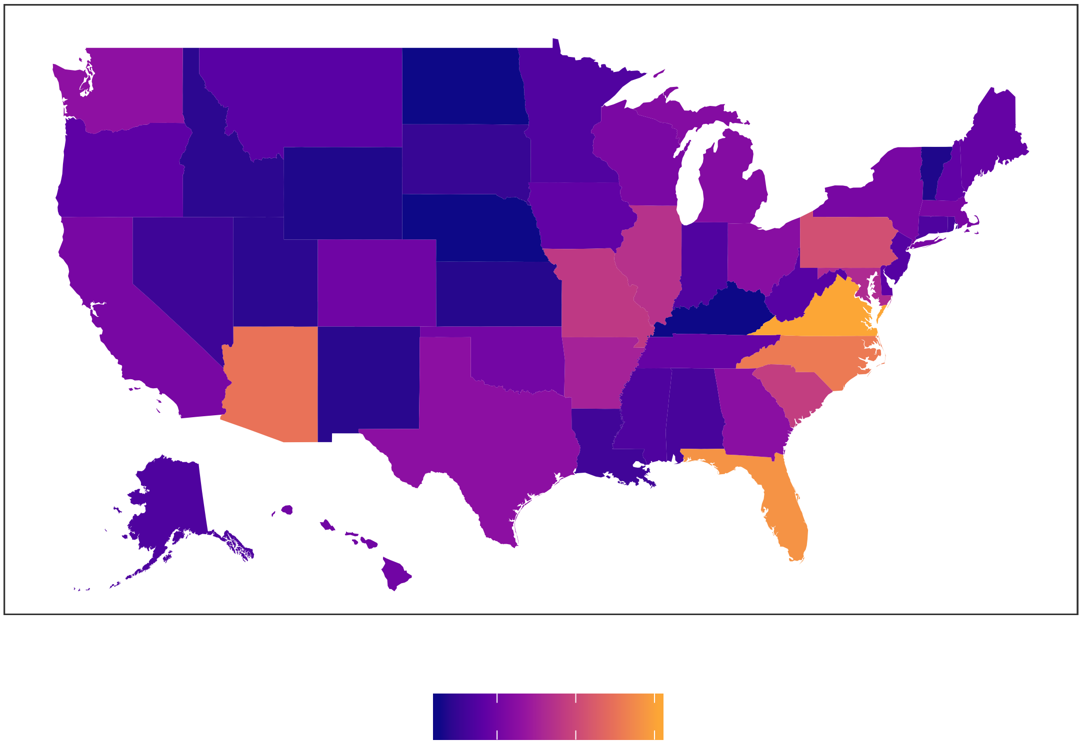
```
d %>%
  filter(chamber == "Senate") %>%
    group_by(member_state, year, pop2010) %>% summarise(n = n()) %>%
    group_by(member_state, pop2010) %>% summarise(mean = mean(n)) %>% ungroup() %>%
    ggplot() +
    geom_point(aes(x = log(pop2010), y = mean), color = "light blue") +
  geom_smooth(aes(x = log(pop2010), y = mean)) +
  geom_text(aes(x = log(pop2010),
                y = mean,
                label = ifelse(mean > mean(mean)*1.5 | mean < mean(mean)*0.5,
                               member_state, "")),
            check_overlap = TRUE,
            size = 2.5,
            hjust = 0) +
  theme_bw() +
  labs(title = "Senator Requests per Year by State Population",
       x = "Log State Population",
       y = "Average Number of Requests per Year")
```

```
## `geom_smooth()` using method = 'loess' and formula 'y ~ x'
```




```
Chamber = "Senate"

d %>%
  ungroup() %>%
  filter(chamber == Chamber) %>%
  group_by(state) %>% summarise(n = n()) %>%
# map_id creates the aesthetic mapping to the state name column
ggplot() +
  # map points to the fifty_states shape data
  geom_map(aes(map_id = state, fill = n), map = fifty_states) +
  expand_limits(x = fifty_states$long, y = fifty_states$lat) +
  coord_map() +
  scale_x_continuous(breaks = NULL) +
  scale_y_continuous(breaks = NULL) +
  labs(x = "", y = "", title = paste("Total Number of Contacts from members of the", Chamber)) +
  scale_fill_viridis_c(option = "C", end = .8) +
  theme(legend.position = "bottom", legend.title = element_blank(),
        panel.background = element_blank()) # + facet_grid(. ~ Constituent)
```

Total Number of Contacts from members of the Senate



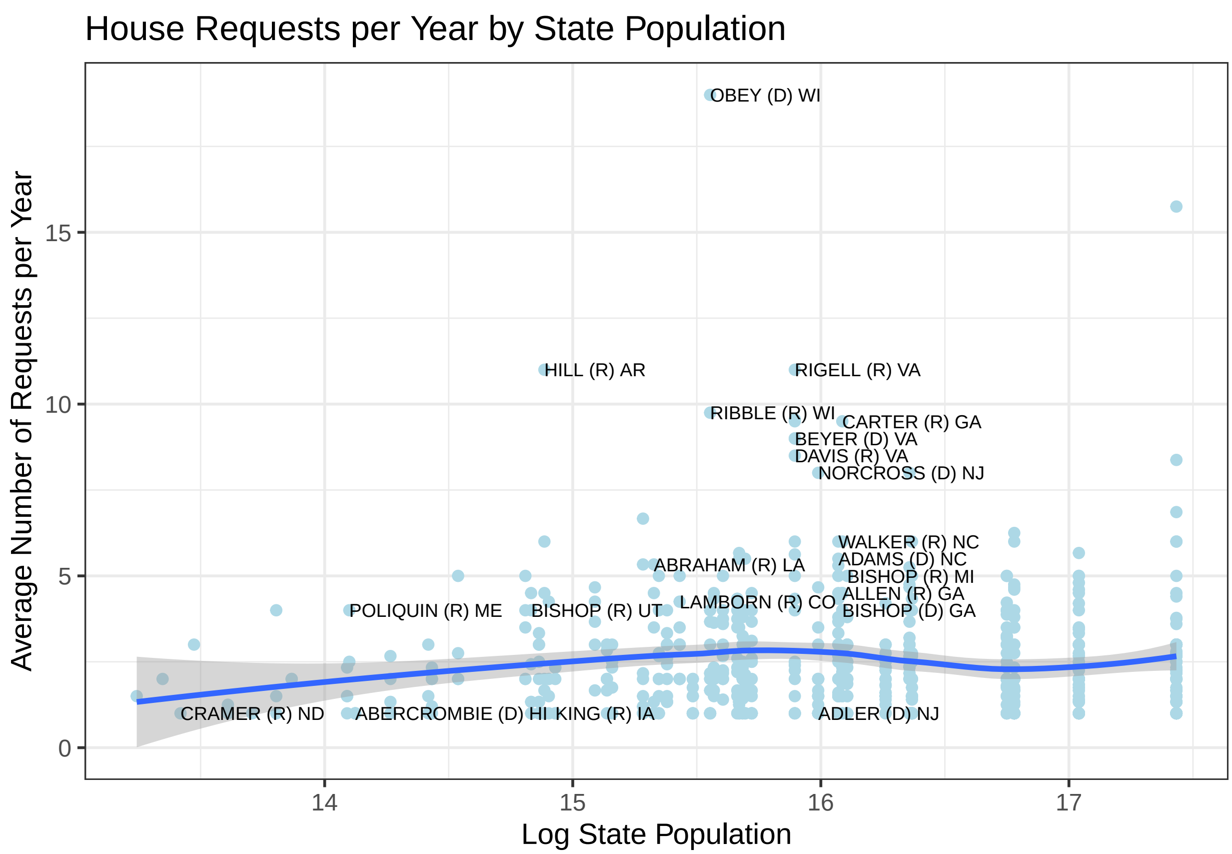

```
##What on earth is happening here?

# d %>%
#   ungroup() %>%
#   filter(chamber == Chamber) %>%
#   group_by(state, pop2010) %>% summarise(n = n()) %>% ungroup() %>%
#   mutate(Per_Capita = n/pop2010*1000000) %>%
#   # map_id creates the aesthetic mapping to the state name column in your data
#   ggplot() +
#   # map points to the fifty_states shape data
#   geom_map(aes(map_id = state, fill = Per_Capita), map = fifty_states) +
#   expand_limits(x = fifty_states$long, y = fifty_states$lat) +
#   coord_map() +
#   scale_x_continuous(breaks = NULL) +
#   scale_y_continuous(breaks = NULL) +
#   labs(x = "", y = "", title = paste("Contacts Per hundred thousand Residents from Members of the", C
#   scale_fill_viridis_c(option = "C", end = .8) +
#   theme(legend.position = "bottom",
#         legend.title = element_blank() )#+facet_grid(. ~ Constituent)
#
# ##Copy. Add constitutent for type 1
#
# d  %>%
#   mutate(Per_Capita = (pop2010)*1000000) %>% ##FIXME non-numeric argument to binary operator
#   # map_id creates the aesthetic mapping to the state name column in your data
#   ggplot() +
```

```
#   # map points to the fifty_states shape data
#   geom_map(aes(map_id = state, fill = Per_Capita), map = fifty_states) +
#   expand_limits(x = fifty_states$long, y = fifty_states$lat) +
#   coord_map() +
#   scale_x_continuous(breaks = NULL) +
#   scale_y_continuous(breaks = NULL) +
#   labs(x = "", y = "", title = paste("Contacts Per hundred thousand Residents from Members of the", C
#   scale_fill_viridis_c(option = "C", end = .8) +
#   theme(legend.position = "bottom",
#         legend.title = element_blank() )+
#   facet_grid(. ~ Constituent)
```

##House Contacts per State ##Contacts per district ##Would this be relevannt??

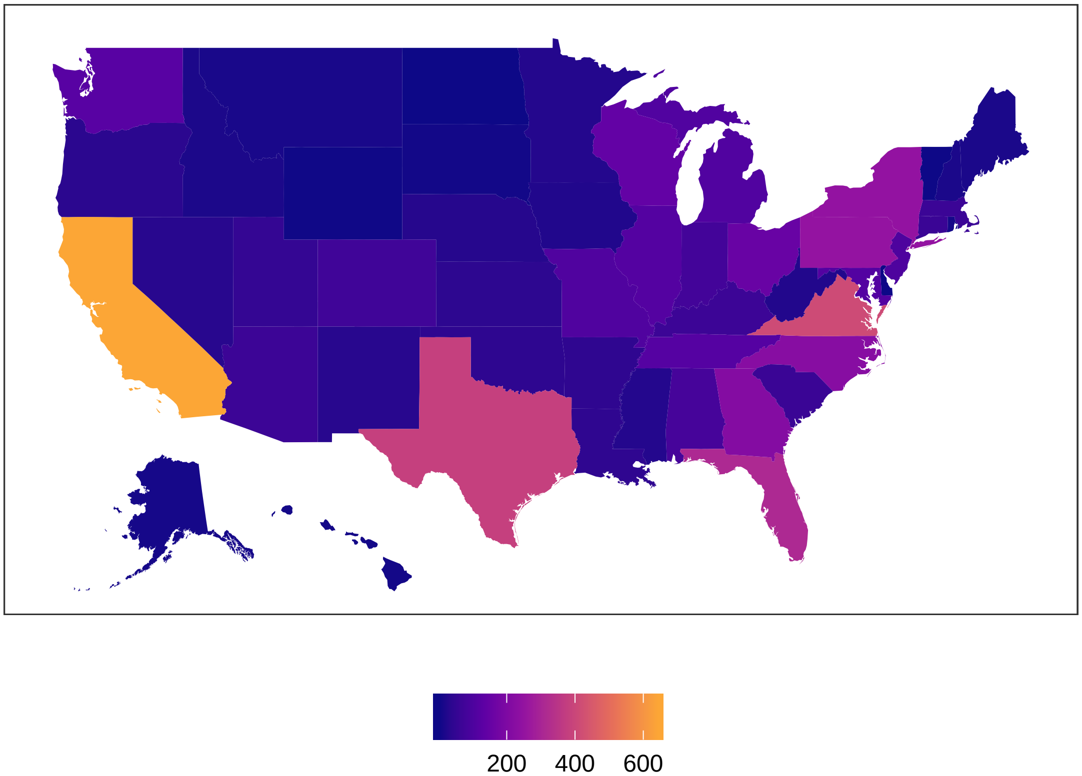
```
d %>%
  filter(chamber == "House") %>%
    group_by(member_state, year, pop2010) %>% summarise(n = n()) %>%
    group_by(member_state, pop2010) %>% summarise(mean = mean(n)) %>% ungroup() %>%
    ggplot() +
    geom_point(aes(x = log(pop2010), y = mean), color = "light blue") +
  geom_smooth(aes(x = log(pop2010), y = mean)) +
  geom_text(aes(x = log(pop2010),
                y = mean,
                label = ifelse(mean > mean(mean)*1.5 | mean < mean(mean)*0.5,
                               member_state, "")),
            check_overlap = TRUE,
            size = 2.5,
            hjust = 0) +
  theme_bw() +
  labs(title = "House Requests per Year by State Population",
       x = "Log State Population",
       y = "Average Number of Requests per Year")
```

```
## `geom_smooth()` using method = 'loess' and formula 'y ~ x'
```




```r
Chamber = "House"

d %>%
  ungroup() %>%
  filter(chamber == Chamber) %>%
  group_by(state) %>% summarise(n = n()) %>%
# map_id creates the aesthetic mapping to the state name column
ggplot() +
  # map points to the fifty_states shape data
  geom_map(aes(map_id = state, fill = n), map = fifty_states) +
  expand_limits(x = fifty_states$long, y = fifty_states$lat) +
  coord_map() +
  scale_x_continuous(breaks = NULL) +
  scale_y_continuous(breaks = NULL) +
  labs(x = "", y = "", title = paste("Total Number of Contacts from members of the", Chamber)) +
  scale_fill_viridis_c(option = "C", end = .8) +
  theme(legend.position = "bottom", legend.title = element_blank(),
        panel.background = element_blank()) # + facet_grid(. ~ Constituent)
```

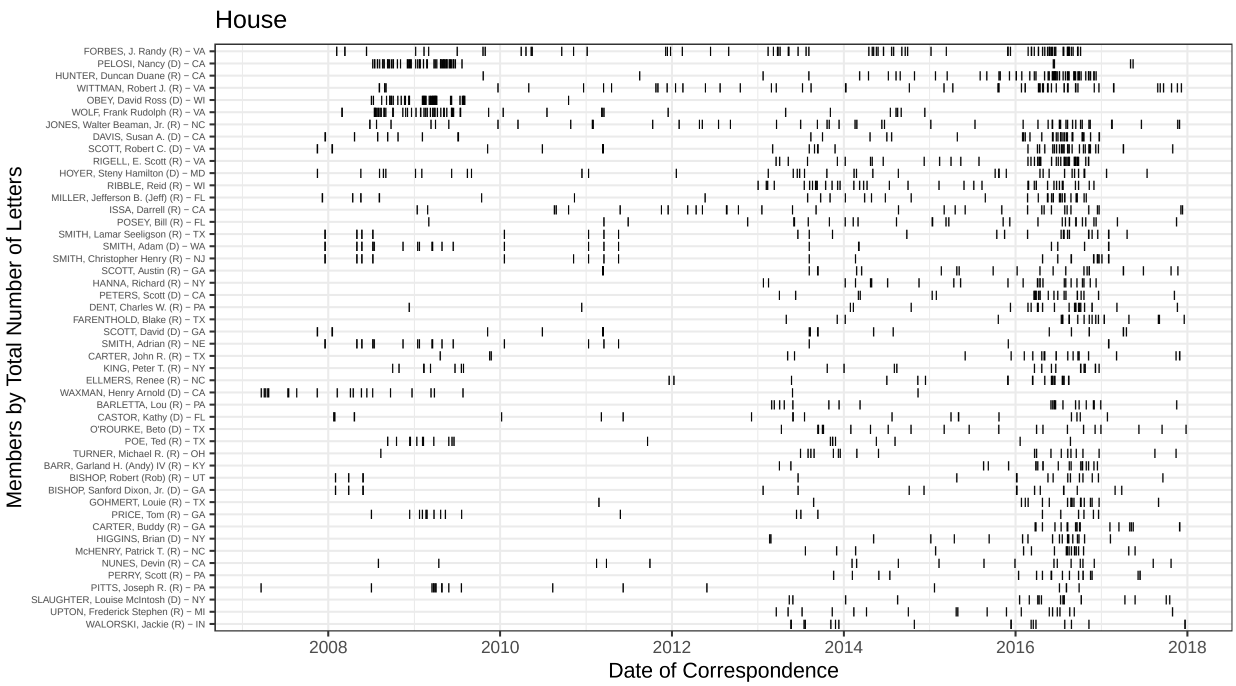
Total Number of Contacts from members of the House



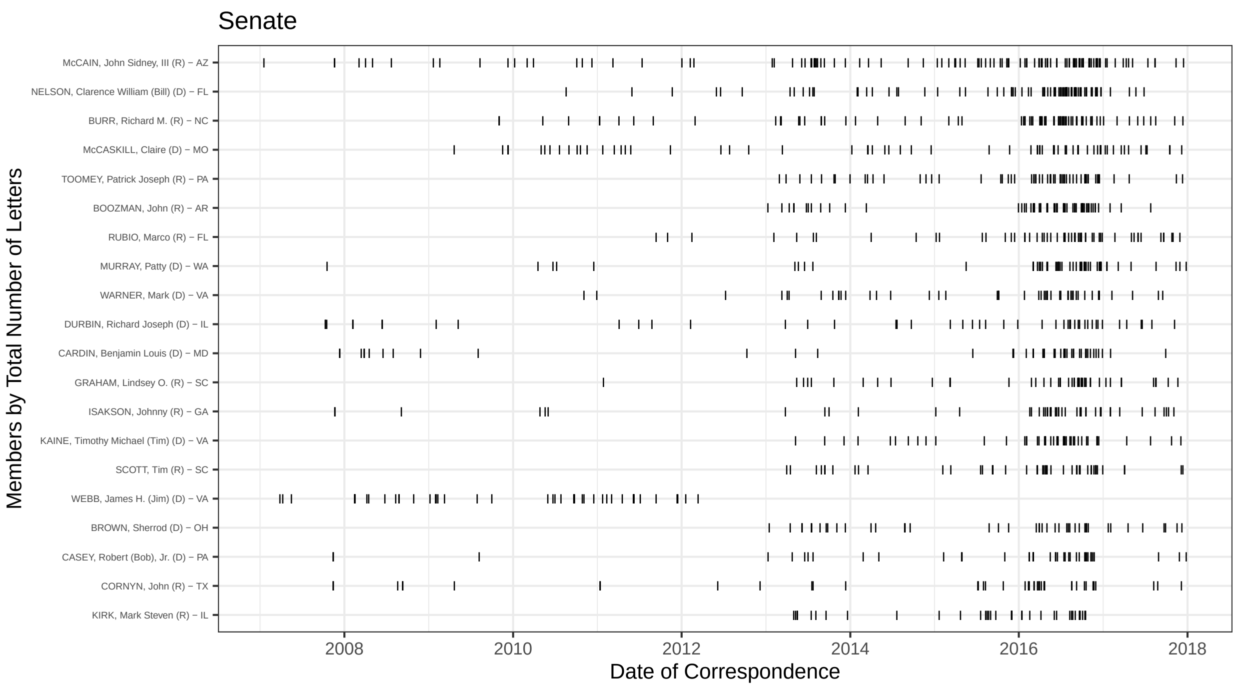
Members who contact DOD more than average:

```
# barcode plot of committee members

# member by year by agency
d %>%
  filter(chamber == Chamber) %>%
  group_by(name_state) %>% mutate(n = n()) %>% ungroup() %>%
  filter(n > mean(n)) %>%
  ggplot() +
  geom_point(
    aes(x = DATE,
        y = reorder(name_state, n) ),
    # alpha = .6,
    shape = 73,
    size=2
  ) +
  labs(title = paste(Chamber),
       y = paste("Members by", "Total Number of Letters"),
       x = "Date of Correspondence" ) +
  theme(
    legend.title = element_blank(),
```

```
    axis.text.y = element_text(size=5)
  ) +
    guides(fill=guide_legend(ncol=1))
```




```
Chamber <- "Senate" # "Senate"

# member by year by agency
d %>%
  filter(chamber == Chamber) %>%
  group_by(name_state) %>% mutate(n = n()) %>% ungroup() %>%
  filter(n > mean(n)) %>%
  ggplot() +
  geom_point(
    aes(x = DATE,
        y = reorder(name_state, n) ),
    # alpha = .6,
    shape = 73,
    size=2
  ) +
  labs(title = paste(Chamber),
       y = paste("Members by", "Total Number of Letters"),
       x = "Date of Correspondence" ) +
  theme(
    legend.title = element_blank(),
    axis.text.y = element_text(size=5)
  ) +
    guides(fill=guide_legend(ncol=1))
```



Members who contact DOD more, relative to other agencies:

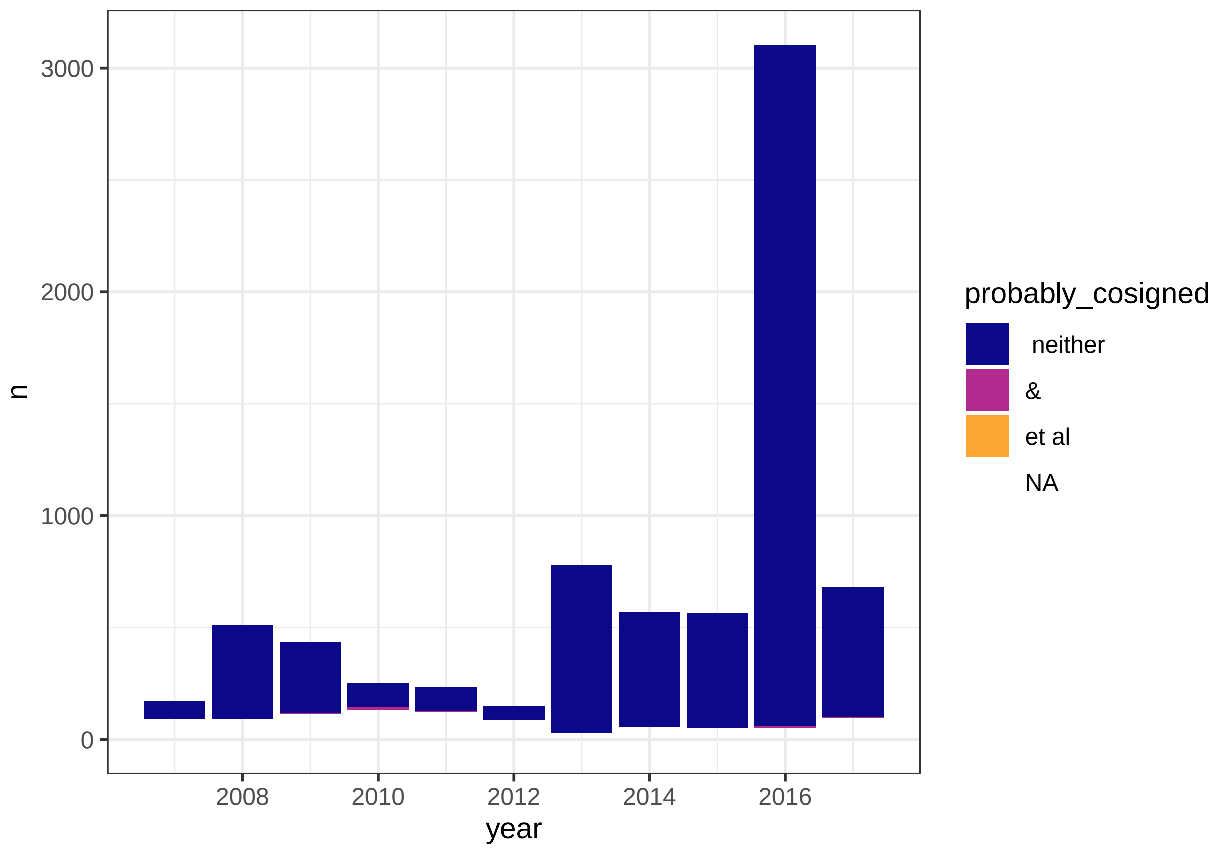
```
# all_contacts %>%
#   filter(congress >109) %>%
#   mutate(DOD = agency == "DOD_OSDJS") %>%
#   count(chamber, OSDJS, name_state, state) %>%
#   spread(key = "OSDJS", value = "n", 0) %>%
#   mutate(ShareToOSDJS = round(`TRUE`/(`TRUE`+`FALSE`),2) ) %>%
#   arrange(-ShareToOSDJS) %>%
#   mutate(total = `TRUE` + `FALSE`) %>%
#   select(name_state, chamber, total, ShareToFERC) %>%
#   filter(ShareToOSDJS>.3) %>%
#   knitr::kable(caption = "Members who contact FERC more, relative to other agencies" )
#
#
# # facet on agency, how does it differ across agencies
#
# chamber <- "Senate" # "Senate" #why does it say False?
#
# all_contacts %>%
#   filter(chamber == "Senate") %>%
#   #filter(str_detect(agency, "DOD")) %>%
#   mutate(DOD = agency == "DOD_OSDJS") %>%
#   group_by(chamber, Department, name_state, state) %>%
#   summarise(n = n() ) %>%
#   ungroup() %>%
#   #spread(key = "DOD", value = n) %>%
#   top_n(10, n) %>%
#   knitr::kable()
```

```
#
#   # mutate(ShareToDOD = round(`TRUE`/(`TRUE`+`FALSE`),2) ) %>%
#   # arrange(-ShareToDOD) %>%
#   # mutate(total = `TRUE` + `FALSE`) %>%
#   # select(name_state, chamber, total, ShareToDOD) %>%
#   # filter(ShareToDOD>.3) %>%
#
#
# chamber <- "House"
#
# all_contacts %>%
#   filter(chamber == "House") %>%  #why is this working this way??
#   mutate(DOD = agency == "DOD$") %>%
#   group_by(chamber, DOD, name_state, state) %>%
#   summarise(n = n() ) %>%
#   ungroup() %>%
#   #spread(key = "DOD", value = n) %>%
#   top_n(10, n) %>%
#   knitr::kable()
#
#  ## Members who contact FERC more in a Congress, relative to other agencies
```

Many letters to DOD are cosigned

#A rough under-estimate based on whether "et al" or "&" appears in the summary: # INVESTIGATE

```
d %>%
  group_by(ID) %>%
  mutate(probably_cosigned = ifelse(str_detect(SUBJECT, "et al"), "et al",
                               ifelse(str_detect(SUBJECT, "&"), "&", " neither"))) %>%
  group_by(year, probably_cosigned) %>%
  tally %>%
  ggplot() +
  aes(x = year, y = n, fill = probably_cosigned) +
  geom_col() +
  scale_fill_viridis_d(option = "C", end = .8)
```



##How to fix this committee Code

#Left join committees into all_contacts #Group by ICPSR and Congress #Summarize committees = paste committee #Full join of all contacts and committees, larger data set

Modify committees variable, ICPSR, Congress, Committee= Armed Services

```
# d %>%
#   mutate(chair = str_remove(chair,"[0-9]* |^NA "),
#          committee = paste(chamber, committees))%>%
#   dplyr::select(committee, position, chair, DATE, assigneddate, terminationdate) %>%
#   distinct() %>%
#   filter(str_detect(committee, "ENERGY$|ENVIRONMENT|NATURAL")) %>%
#   mutate(position = ifelse(position == "Other", NA, position)) %>%
#   #filter(# !is.na(position),chair_since_2007 == T) %>%
#   drop_na(chair) %>%
#   dplyr::select(DATE, chair, position, assigneddate, terminationdate, committee) %>%
#   distinct() %>%
#   ggplot() +
#   geom_point(
#     aes(x = DATE,
#         y = chair),
#     shape = 73,
```

```r
#     size=2
#   ) +
#   geom_segment(aes(y = chair, yend = chair,
#                    x = date, xend = date,
#                    linetype = factor(position)),
#                position = position_nudge(y = -0.3)) +
#   labs(title = paste("Letters to FERC from Committee Leadership"),
#        y = "",
#        x = "",
#        linetype = "Leadership position") +
#   scale_y_discrete(position = "right") +
#   theme(
#     strip.text.y = element_text(angle = 180, size = 5),
#     # legend.title = element_blank(),
#     axis.text.y = element_text(size=5),
#     axis.text.x = element_text(angle = 0)
#   ) +
#   facet_grid(committee ~ ., scales = "free_y", space = "free_y", switch = "both")
```

```r
# MEMBER DEMOGRAPHICS


# gender for those where we have the data from LEP # WE HAVE BETTER DATA, NEEDS TO BE MERGED IN
d %<>%
  # merge LEP data into df
  left_join(
    # read in the LEP data
    read.csv(here("members/LEP111to113.csv")) %>%
      # just grabbing female variable for now
      select(icpsr, female) %>%
      # distinct icpsr-gender combinations
      distinct() %>%
      #make ICPSR numbers numeric to merge with df
      mutate(icpsr = as.numeric(icpsr))
  )
```

```
## Joining, by = c("icpsr", "female")
```

```
## Warning: Column `female` joining character vector and factor, coercing into
## character vector
```

Limitations throughout the Independent Study

Throughout the course of this independent study, there were a few challenges and limitations I faced.

Future ideas to explore with this paper

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#Notes:

1. Count letters by member by year (including zero counts)
2. Policy Stuff
3. Divide by years–annualized by years
4. Control with Gender–how long they're in congress
5. Veterans–More likely to write, shared Veteran status. Veterans-gender-bias
6. What Kind of Variate Models??–Pro-business/anti-business/Constituent vs. everyone else/ Party Explanatory #Party # constituent vs non constituent
7. Best way to present paper..like this or should I hide code?? #HTML
8. Electoral Interests?–how to measure that? #Most important issue data #State